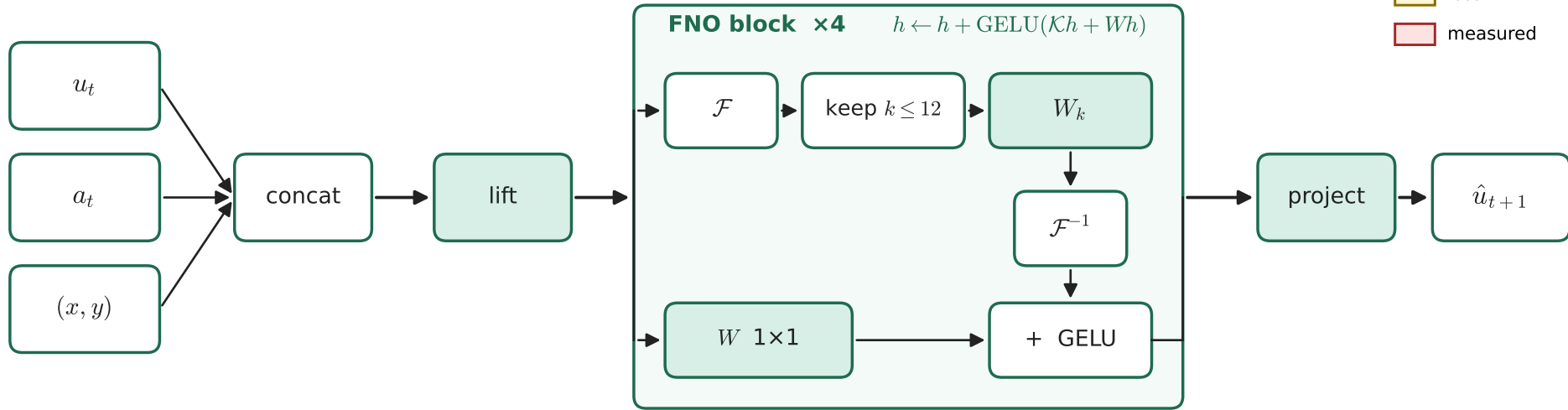


SIMULATE — differentiable surrogate G_θ

Lipschitz mode: $\|W_k\|_2 \leq 1$, L_G measured



$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_p \mathcal{L}_{\text{phys}} + \lambda_r \mathcal{L}_{\text{roll}}$$

$\mathcal{L}_{\text{data}}$

$$\mathcal{L}_{\text{phys}} = \|\dot{\hat{u}} - \text{rhs}(\hat{u}, a)\|$$

$\mathcal{L}_{\text{roll}}$
H steps

sharp fronts: $\ell = \ell_{\text{UNet}} + g \cdot \ell_{\text{FNO}}$, gate g learned